

Context-Dependent Selection of Urban Routing Policies: Regime Effects, Decision Value and Generalisation Limits

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Abstract. Route guidance systems normally commit to one routing objective before any trip is planned, even though the relative merit of established objectives can depend on the traffic state. This paper studies the selection of the routing *policy* rather than of a route. Using 432 completed SUMO simulations over 24 controlled contexts on a twelve-signal urban mesh, we evaluate shortest-distance routing (SP), dynamic travel-time routing (DTT) and constrained travel-time-aware eco-routing under three demand levels, four restriction configurations and two signal plans, with preference expressed as a fixed-reference additive cost over mean journey time and tailpipe CO₂. SP has the lowest observed cost in all eight low-demand contexts and DTT in all eight high-demand contexts, and the signal plan displaces the crossing by at least one step of the demand grid. Relative to the single best fixed policy, leave-one-context-out selection lowers mean decision regret from 0.03018 to 0.00606 normalised cost units — 79.9 % of the available decision headroom captured, with the lowest-cost policy chosen in 21 of 24 contexts. A two-threshold rule on demand and signal plan reproduces that result exactly, so the benefit follows from the regime structure rather than from model capacity. A leave-one-demand-out test reverses the conclusion: pooled regret rises to 0.07649, above the fixed-policy benchmark. Policy selection therefore has measurable value inside the regimes represented during fitting and can be worse than a fixed policy outside them. The contribution is the empirical characterisation of this decision boundary and its limits; no new routing, multi-criteria or learning algorithm is proposed.

Keywords: Routing-policy selection · Urban traffic simulation · Algorithm selection · Decision regret · Traffic regimes · Generalisation.

1 Introduction

A route guidance system makes two decisions that are easy to conflate. The first is the classical route-choice problem: which path should a vehicle take from an origin to a destination? The second is rarely treated as a decision in its own right: which routing *policy* should generate that path? An operator may minimise static distance, respond to recently observed travel times, or minimise estimated

emissions subject to a travel-time budget. Each policy returns a different path for the same request, so the choice of policy precedes and constrains the choice of route. This paper studies the second decision.

The distinction matters because congestion changes the relative value of routing objectives. On a lightly loaded network the shortest path is usually also fast, and minimising distance gives up little. As load rises, repeated concentration of traffic on the shortest corridor raises delay there faster than on the alternatives, and a policy that reacts to observed travel times can become preferable. Signal timing modifies the same relationship, because the green split determines how much capacity each competing movement receives. These mechanisms are individually well documented at the level of routes and traffic dynamics, but they do not answer the question an operator faces: should one routing objective be fixed for the whole day, or should the active objective change with the operating context, and how much is that change worth?

We treat this as a context-dependent algorithm-selection problem in the sense of Rice [24]. The object being selected is a routing policy, not a route: each policy remains responsible for assigning paths according to its own objective, and a context-aware layer only decides which policy is active for the current operating episode. Separating the two levels makes it possible to measure the value of adapting the policy without attributing it to any particular route-generation mechanism.

The central research question is:

Under which traffic regimes does the preferred established routing policy change, how much decision value is available from adapting that policy, and how reliably can the resulting structure be identified on unseen contexts?

Three subsidiary questions follow. How does the signal plan move the demand range in which the preferred policy changes? How much of the available decision headroom can a simple interpretable rule recover on held-out contexts, and does a flexible model recover more? And what happens when the rule is applied outside the range of operating conditions represented while it was fitted?

The paper makes four empirical contributions on a benchmark of 432 completed simulations. It characterises a regime-dependent preference structure among established routing policies on a signalised network, and shows that the signal plan displaces the switching range by at least one step of the tested demand grid. It quantifies the value of adaptation with context-level decision regret against a hindsight benchmark rather than with prediction error, and reports how that value is distributed across regimes. It shows that a two-threshold rule on demand and signal plan attains the same held-out decision quality as a fitted regression tree, which bounds what the learned model contributes. And it identifies an explicit generalisation boundary: the procedure that helps inside the represented regime space becomes worse than the fixed policy it replaces when an entire demand regime lies outside it. No new routing algorithm, multi-criteria method or learning method is proposed, and no claim of priority is made.

2 Literature Review

The relevant work spans three strands. Each is summarised below in terms of what it establishes; the third closes with the position of the present study.

2.1 Dynamic and Traffic-Aware Routing

Shortest-path computation on a static network is the classical starting point [7]. Dynamic routing replaces static link attributes with time-dependent or observed travel times, and stochastic formulations treat network conditions as random and solve for routing policies rather than fixed paths [11,17]. Proactive rerouting distributes vehicles over alternatives to relieve anticipated congestion, and stochastic optimal-control formulations extend this to uncertain demand and heterogeneous behaviour [21,23].

A second body of work establishes that the benefit of traffic information depends on the operating state. Empirical studies of real-time rerouting report larger gains during congested periods and for trips whose alternatives materially change experienced travel time [10]; simulation and analytical studies identify demand-dependent transitions in the effectiveness of dynamic guidance, including regimes in which rerouting ceases to help [19,5,6]; and work on route choice under different control systems shows that the control regime itself can change the relative performance of route-choice strategies [27,22]. What this literature does not establish, and what is measured here, is where the *preference order* between established policies reverses as a joint function of demand and signal plan, and what an operator gains by acting on that reversal rather than by improving any single mechanism.

2.2 Environmental, Multi-Criteria, and Context-Dependent Routing

Environmental routing incorporates emissions or fuel use into path costs using historical or real-time traffic information [2,29]. Because a minimum-emission path can be much slower than the fastest path, emission objectives are commonly constrained by a travel-time budget, producing the constrained eco-routing family used here [30,12]; vehicle-specific formulations show further that the relationship between route choice, traffic state and emissions depends on vehicle and network characteristics [26].

The multi-criteria literature supplies established ways of representing preferences over competing outcomes, including additive value models and outranking or compromise methods [14,13,20,3,8]. Here such a model is used only to make “preferred policy” operational, not as a contribution, and the check it implies — reported in Section 6.9 — is whether the chosen criteria discriminate among the candidate policies at all, since a preference model that changes no decision adds nothing however sophisticated. What is measured here is whether the two primary criteria are sufficiently independent in a signalised urban setting for a preference model to matter, and whether a third, system-level congestion criterion changes which policies are mutually incomparable.

2.3 Algorithm Selection and Decision-Oriented Evaluation

The algorithm-selection framework formalises the observation that different algorithms are preferable on different problem instances, and that instance features can be mapped to expected performance [24]; portfolio systems operationalise this and benchmark libraries standardise how they are evaluated [28,1,15], while instance-space analysis shows how performance varies across the problem space [25]. Two reference points from that literature are used throughout: the single best fixed algorithm, the strongest deployable non-adaptive choice, and the virtual or hindsight best, which selects per instance using knowledge available only after the fact. A second relevant distinction is between predictive accuracy and decision quality: a model may predict poorly yet decide well when the ranking of alternatives is preserved, while a small prediction error can be costly if it changes the selected alternative [9]. This motivates evaluating a selector by decision regret against the per-instance best [16] rather than by regression error.

Position of this study. The three strands meet at a question none answers alone. The routing literature establishes that policy performance is state-dependent but compares mechanisms rather than deciding among them; the multi-criteria literature supplies preference models but does not test whether they discriminate in a given environment; the algorithm-selection literature supplies the decision-theoretic apparatus but has not been applied to routing policies under controlled traffic regimes. This study occupies that intersection and contributes no new mechanism — the policies are established objectives, the preference model is a standard fixed-reference additive cost and the selector is a shallow regression tree. What is new is the measurement: a controlled, fully replicated characterisation of when the preferred routing policy changes, what that change is worth in decision terms, how little model capacity is needed to exploit it, and where the resulting rule stops being valid.

3 Problem Formulation

3.1 Contexts, policies and the two-level decision

Let $\mathcal{S}$ be a set of operating contexts and $\mathcal{A}$ a finite set of retained routing policies. A context $s \in \mathcal{S}$ contains only quantities known *before* the policy is activated; here it is the triple (scheduled primary demand, planned restriction configuration, signal-control plan), and no realised travel time, queue, flow or emission is part of it. A policy $a \in \mathcal{A}$ maps a route request to one path from a finite admissible catalogue according to its own objective.

The decision has two levels: *Level 1*, policy selection, maps the context s to one policy $a \in \mathcal{A}$; *Level 2*, route assignment, is performed by that policy according to its own objective. Only Level 1 is learned. The unit of analysis is the context, not the vehicle and not the replication: a context is one instance in the algorithm-selection sense, all of its replications move together through every evaluation protocol, and the selector never observes the realised outcome of any candidate policy in the context it decides about.

3.2 Decision cost, regret and headroom

Let y_{sak} be the mean outcome of criterion k in context s under policy a , averaged over replications after run-level aggregation. With positive fixed references r_k and weights $w_k \geq 0$, $\sum_k w_k = 1$, the scalar decision cost is

$$C_{sa} = \sum_k w_k y_{sak} / r_k, \quad (1)$$

lower being preferred. The references are fixed for the whole benchmark rather than recomputed per context, so costs are comparable across contexts and a context in which every policy is slow is not rescaled into apparent equality; one cost unit is one benchmark-average criterion value.

The best policy actually observed in a context is $a_s^* \in \arg \min_{a \in \mathcal{A}} C_{sa}$, with cost C_s^{VB} . Selecting a_s^* everywhere defines the *hindsight* (virtual best) policy: a bound, not a method, since it uses the realised performance of every candidate. For a selection rule $\pi : \mathcal{S} \rightarrow \mathcal{A}$ the *decision regret* $R_s(\pi) = C_{s,\pi(s)} - C_s^{\text{VB}} \geq 0$ measures the cost of the decision, not the error of a prediction. The strongest deployable non-adaptive rule is the *single best fixed policy* $a^{\text{SB}} = \arg \min_a |\mathcal{S}|^{-1} \sum_s R_s(a)$, and the quantity adaptation can win is the *decision headroom* $\bar{H}$, of which a rule recovers the fraction $G(\pi)$:

$$\bar{H} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} (C_{s,a^{\text{SB}}} - C_s^{\text{VB}}), \quad G(\pi) = \frac{\bar{H} - \overline{R(\pi)}}{\bar{H}}. \quad (2)$$

Thus $G = 1$ for the hindsight bound, $G = 0$ for the single best fixed policy, and $G < 0$ for a rule worse than doing nothing adaptive. Because $\bar{H}$ is defined on a finite context set, G is a benchmark-specific decision statistic. It is *not* a percentage reduction in travel time, emissions or any other physical quantity; the two are reported separately throughout.

3.3 Three properties that must not be confused

Complementarity asks whether different policies win in different contexts. It is a property of the portfolio and environment alone, established by observing that a_s^* is not constant over $\mathcal{S}$, and it says nothing about how much the variation is worth. *Decision value* asks how much cost the variation carries, and is $\bar{H}$ in (2); a portfolio can be strongly complementary while $\bar{H}$ is negligible, if the single best fixed policy is nearly optimal wherever it is not optimal, so complementarity is necessary for decision value but not sufficient. *Deployability* asks whether the variation can be exploited from pre-decision information on contexts not used for fitting, and is $G(\pi)$ under a held-out protocol together with evidence about the range of contexts over which the rule remains valid; positive decision value does not imply it, and a rule deployable on one region of the context space need not be deployable outside it. All three are measured separately in Section 6, and this study reports a different answer for each.

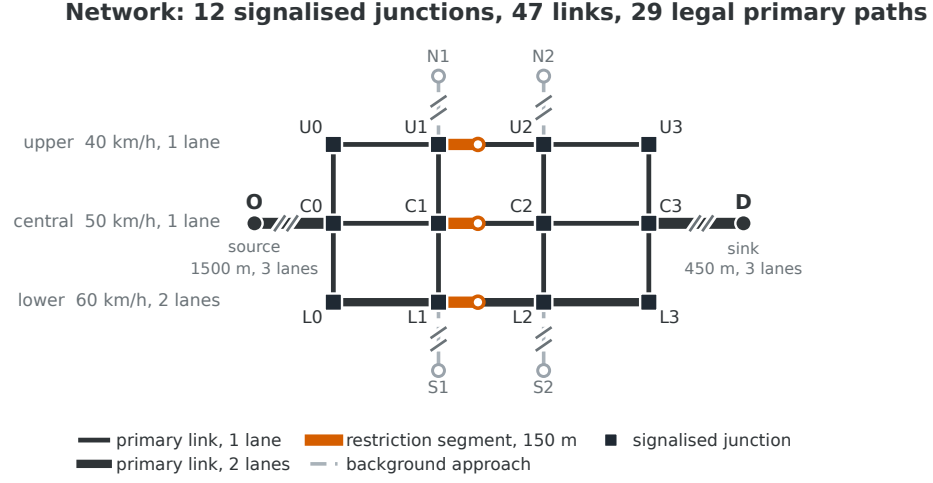


Fig. 1. The fixed network. Line weight encodes lane count, the three alternative restriction segments are highlighted, and background approaches are dashed with their stubs shortened at the break marks. Exactly one restriction segment is active in a restricted context. The 24 contexts built on this network are shown in Fig. 3b.

4 Methodology

4.1 Simulation Environment and Network

All experiments use one fixed topology: a bidirectional 3×4 signalised mesh with 12 signalised junctions, 47 external links and 29 legal simple primary paths (Fig. 1). The upper, central and lower corridors have speed limits of 40, 50 and 60 km/h and one, one and two lanes, so the three east–west corridors differ in both free-flow speed and capacity, and three alternative eastbound locations carry a 150 m speed-restriction segment with a speed ratio of 0.35. Signals run a common 60 s cycle: the balanced plan allocates approximately 26 s each to the competing east–west and north–south movements, the arterial-priority plan 36 s and 16 s, with identical clearance intervals. Four background streams load the north–south movements that compete with the primary corridors, and a three-lane source and sink ensure that neither end of the corridor is the binding constraint. Lane counts are specified design inputs, not calibrated saturation flows, which limits the external interpretation of absolute values. Simulation uses SUMO [18] with teleporting disabled, so a vehicle that cannot be inserted waits rather than disappearing from the accounting. Table 1 lists the configuration.

4.2 Routing Policy Portfolio

Let $\mathcal{P}$ be the 29 admissible paths, $L(p)$ the static length of path p , and $\hat{T}_t(p)$, $\hat{E}_t(p)$ the travel-time and emission estimates available at decision time t . The

Table 1. Simulation configuration. Every item is common to all 432 runs; the factors that vary between contexts are listed in Section 5.1.

Item	Setting
Network	Bidirectional 3×4 mesh; 12 signalised junctions; 47 external links; 29 legal simple paths; 400 m horizontal and 300 m vertical spacing
Corridors	Upper 40 km/h, 1 lane; central 50 km/h, 1 lane; lower 60 km/h, 2 lanes
Signal cycle	60 s, with two 3 s yellow and 1 s all-red clearance intervals
Background demand	Four north–south streams, 200 veh/h each
Simulator and fleet	SUMO, 1 s step, teleporting disabled; homogeneous fleet, HBEFA3 PC_G_EU4
Restriction segment	150 m, speed ratio 0.35, in one of three alternative eastbound locations
Signal plans	balanced 26/26 s; arterial-priority 36/16 s
Measurement window	600 s warm-up; primary requests generated to 2400 s and followed to arrival

implemented objectives are

$$p_{\text{SP}} \in \arg \min_{p \in \mathcal{P}} L(p), \quad p_{\text{DTT}} \in \arg \min_{p \in \mathcal{P}} \hat{T}_t(p), \quad (3)$$

$$p_{\text{ECO}} \in \arg \min_{p \in \mathcal{P}} \hat{E}_t(p), \quad p_{\text{TECO}, \alpha} \in \arg \min_{\{p: \hat{T}_t(p) \leq (1+\alpha) \min_q \hat{T}_t(q)\}} \hat{E}_t(p), \quad (4)$$

with $\alpha \in \{0.05, 0.10, 0.15\}$. These are finite-candidate implementations of established objectives [7,2,30,12], not new routing algorithms, and the three TECO settings are parameter values of one family rather than three principles.

Table 2. Routing policies, their objectives and their role in the analysis.

Policy	Objective	Role
SP	Minimise static path length	Decision portfolio
DTT	Minimise estimated travel time on the latest observed interval	Decision portfolio
TECO10	Minimise estimated CO ₂ subject to a 10 % travel-time budget	Decision portfolio
TECO05, TECO15	The same objective, 5 % and 15 % budgets	Allowance sensitivity; measured, outside the portfolio
ECO	Minimise estimated CO ₂ , no time budget	Negative control; removed by a route-ordering gate applied before any decision analysis

Every policy uses the same network, demand manifest, signal plan, fleet and admissible catalogue, so the only difference between two runs of one context is the objective used to pick a path. Estimates come from completed 60 s observation intervals, so a request at time t can see neither the future nor the requesting vehicle’s own realised trip; each vehicle receives one route at departure and does not re-route; compliance is complete. The exclusion of ECO is a pre-decision screening step reported in Section 6.9: its emission forecasts were accurate in aggregate yet ranked routes badly, so it was removed before the portfolio was analysed and retained as an audited negative control. The decision portfolio is therefore $\mathcal{A} = \{\text{SP}, \text{DTT}, \text{TECO10}\}$.

4.3 Decision Criteria

Two criteria are primary: mean journey time of the measured cohort, from requested departure to arrival, and mean tailpipe CO_2 per vehicle of the same cohort. Journey time therefore includes insertion delay incurred when the network cannot immediately accept a vehicle, whereas in-network CO_2 excludes pre-insertion idling because a vehicle that has not entered emits nothing in the model. This asymmetry is a measurement boundary rather than an error, and its effect on every decision is measured in Section 6.9. Fuel is excluded: in this homogeneous fleet it is proportional to CO_2 to within numerical precision.

Network stopped vehicle-time is a congestion diagnostic rather than a primary criterion, because it is accumulated over *all* vehicles including background traffic and is therefore a system-level quantity rather than a service level experienced by the measured cohort; promoting it after the fact would silently change what “preferred policy” means. The primary two-criterion rule is used unchanged for every regret and headroom quantity in this paper, and stopped vehicle-time enters a separate, explicitly labelled multi-criteria analysis in Section 6.8. Route distance and the 95th percentile of journey time are likewise diagnostics. Equal weights are used for the two primary criteria, and the references r_k are their grand means over all contexts and portfolio policies: 542.844 s and 1263.515 g.

4.4 Decision Cost and Regret

Criterion means are computed within each run and then averaged over the three paired replications of a context before Eq. (1) is applied, so a context contributes one cost per policy. The seeds are paired across policies — seed j drives the same exogenous demand realisation whichever policy is active — so a difference between two policies in one context is a paired difference. Two conventions affect the reported values: the references r_k are computed once on the full benchmark for scoring, while in every held-out protocol the model and its internal scales are refitted on training contexts only (Section 6.9 reports what happens when the scoring scales are restricted too); and the best fixed policy is identified inside each fold, so the fixed baseline is itself a fitted quantity rather than an oracle.

4.5 Context-Aware Policy Selection

The selector predicts the criterion outcomes of every portfolio policy from the context and then applies Eq. (1) to the predicted outcomes. Its inputs are the scheduled demand, the planned restriction configuration and the signal plan; no realised outcome of any kind enters the feature vector, and all features are known before any vehicle departs.

Rather than reporting one model, we evaluate a ladder of rules of increasing capacity under the same held-out protocol, listed in Table 5: the single best fixed policy identified on each fold’s training contexts (R0), which adaptation must beat; a mechanistic demand threshold choosing SP below a fitted value and DTT above it (R1) and the same rule with one threshold per signal plan (R2); multi-output regression trees of depth 1 (R3) and of depth 3 with minimum leaf size 2, the pre-registered setting [4] (R4); and the hindsight best as an upper bound (R5). R1 and R2 are explicit rules whose only free parameters are demand thresholds fitted by minimising training regret, so including them attributes any advantage of R4 to model capacity rather than to the regime structure itself; the comparison is reported whichever way it falls.

4.6 Validation and Robustness

The primary held-out protocol is leave-one-context-out: for each held-out context the model, its reference scales and the identity of the best fixed policy are refitted from the remaining 23 contexts, and all three replications of the held-out context stay together. Three further protocols probe different failure modes — leave-one-demand-out, which removes an entire demand level and therefore tests behaviour outside the fitted support; seed-block evaluation, fitting on two seeds and scoring on the third; and a training-only scaling variant, which removes the last dependence of the score on held-out data. The sensitivity checks and the two negative controls are listed in Section 5.4 and reported in Table 6.

5 Experiments

5.1 Experimental Design

The design is a full factorial of three primary demand levels (720, 1200 and 1680 veh/h), four restriction configurations (none, upper, central or lower corridor) and two signal plans (balanced and arterial priority), giving 24 contexts. Each is simulated under six policy settings with three paired seeds, so the campaign consists of $24 \times 6 \times 3 = 432$ runs. Demand levels are separated by 480 veh/h, and that spacing determines what the experiment can resolve about the location of a preference change (Section 5.3).

Four experiments are run on this campaign, ordered so that no later result can redefine an earlier procedure. **E1** measures the switching margin between SP and DTT in every context and determines, per restriction and signal-plan cell, the demand interval in which it changes sign; **E2** compares the fixed policies, the

ladder R0–R4 and the hindsight bound using context-level regret; **E3** withholds entire demand levels to separate interpolation inside the fitted support from extrapolation beyond it; and **E4** runs the sensitivity checks and the two negative controls.

5.2 Main Simulation Campaign

All 432 runs completed and were retained; no run was discarded, re-run or replaced, and no cell is missing. Accounting validity was checked before any decision analysis: the largest discrepancy between edge-accumulated and trip-accumulated emission mass is 1.69×10^{-3} against a 10^{-2} tolerance, and the largest relative error of the native emission algebra is 6.9×10^{-7} against a 10^{-4} tolerance. A separate route-probe campaign of 232 runs — four contexts, two seeds and all 29 admissible paths driven explicitly — screened the emission forecast that the ECO and TECO policies consume; it is the source of the decision to exclude ECO and is reported with the negative controls in Section 6.9.

5.3 Switching-Boundary Resolution

The experiment measures the switching margin at three demand levels per cell. When the margin changes sign between adjacent levels it identifies the *interval* containing the change, not a threshold inside that interval, and no interpolated threshold is reported anywhere in this paper. What the completed data can establish is whether each bracket is itself resolved — whether the observed sign of the margin at each endpoint is large relative to dispersion across paired replications — and that analysis, in Section 6.3, requires no additional simulation.

Localising the change would require simulating intermediate demand levels on this same network: two additional levels across the eight cells and three portfolio policies, at three seeds, is 144 runs. Those runs were not executed. The network and scenario configuration files for this benchmark are not present in the analysis environment, and a reconstructed network would not be comparable with the 432 completed runs, so the additional runs could not be produced on the same footing as the existing evidence. No claim here depends on them: every statement about the switching location is a statement about an interval of the tested demand grid.

5.4 Sensitivity and Robustness Experiments

Fifteen combinations of tree depth $\in \{1, 2, 3, 4, \text{unlimited}\}$ and minimum leaf size $\in \{1, 2, 3\}$ are evaluated under the primary protocol; the time weight is swept over $\{0, 0.25, 0.5, 0.75, 1\}$; the three seed blocks are each held out in turn; the scoring scales are recomputed from training contexts only; and every decision is recomputed with journey time replaced by in-network time, with the tail criterion in place of the mean, and with route distance and then network stopped vehicle-time added as a third criterion. The two negative controls are the ECO route-ordering gate and the weight sweep, both designed to fail visibly if the corresponding component is doing no work.

5.5 Statistical Treatment

Each context is replicated three times with paired seeds, so a comparison between two policies in one context is a paired difference d_j , $j = 1, 2, 3$. We report the mean margin $\bar{d}$ and the standard error $\text{SE}(\bar{d}) = s_d/\sqrt{3}$, and call a margin *resolved* when $|\bar{d}| > 2\text{SE}(\bar{d})$. This criterion is descriptive: it marks which observed orderings are larger than the replication dispersion that produced them. It is **not** a test of statistical significance — with three replications the standard error rests on two degrees of freedom, no distributional assumption is made, no null hypothesis is specified and no multiplicity correction is applied. The paper contains no hypothesis test and makes no claim of statistical significance anywhere. Aggregate quantities such as mean regret are reported as benchmark statistics, and where their sensitivity to replication matters the seed-block analysis of Section 6.9 shows the spread directly rather than substituting an interval for it.

6 Results and Discussion

6.1 Campaign Validity

All 432 runs completed and are used; each of the 144 context–policy cells contains three paired replications, and the measured cohort ranges from 361–392 vehicles per run at 720 veh/h to 862–906 at 1680 veh/h, totalling 270,432 trips.

One property of the campaign shapes everything that follows. Insertion delay — time spent waiting to enter the network — is negligible at 720 veh/h (0.12 s averaged over all 432 runs), small at 1200 veh/h (5.54 s) and large at 1680 veh/h (262.41 s, with 810.29 s in the worst single run), and it is strongly policy-dependent: at 1680 veh/h it averages 447.5 s under SP against 6.3 s under DTT. The highest demand level therefore operates at the edge of what the network can accept, and absolute journey times there partly describe the insertion queue rather than in-network travel. Section 6.9 shows that the policy *ordering* does not depend on this boundary; the absolute values clearly do.

6.2 Regime-Dependent Policy Preference

Define the switching margin between the two unconstrained policies as

$$m_s = C_{s,\text{SP}} - C_{s,\text{DTT}}, \quad (5)$$

so a negative value means SP has the lower observed cost. Fig. 2 shows the physical outcomes, Fig. 3 the resulting structure and Table 3 the values.

The structure is systematic rather than confined to one configuration: at 720 veh/h the margin is negative in all eight cells and at 1680 veh/h positive in all eight. Over the full portfolio, SP has the lowest observed cost in all eight low-demand contexts and DTT in all eight high-demand contexts, while the intermediate level splits three ways — SP in three contexts, TECO10 in three

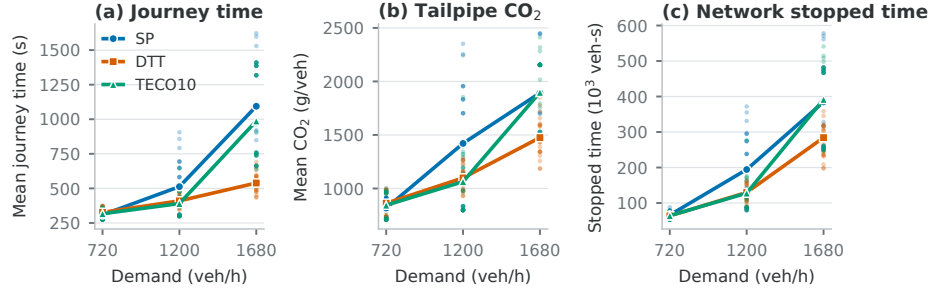


Fig. 2. Policy outcomes against demand. Lines are means over the eight contexts at each demand level; faint points are individual runs. The three panels separate the two primary criteria from the system-level congestion diagnostic. The policies are near-indistinguishable at 720 veh/h and separate sharply above it.

Table 3. Switching margin m_s with its standard error over three paired seeds. Negative values favour SP. The bracket is the interval of the tested demand grid in which the sign changes; it is not a threshold estimate. $\dagger$ marks the two contexts where $|m_s| \leq 2 \text{ SE}$.

Signal	Restr.	720 veh/h		1200 veh/h		1680 veh/h		Bracket
		m_s	SE	m_s	SE	m_s	SE	
Bal	none	−0.0551	0.0064	+0.4204	0.0334	+1.0139	0.0323	(720, 1200]
Bal	upper	−0.0515	0.0071	+0.4382	0.0697	+0.9418	0.0232	(720, 1200]
Bal	central	−0.0195	0.0220 [†]	+0.8056	0.0502	+1.1137	0.0233	(720, 1200]
Bal	lower	−0.0562	0.0058	+0.4196	0.0334	+1.0134	0.0323	(720, 1200]
Art	none	−0.0184	0.0046	−0.1269	0.0101	+0.3249	0.0162	(1200, 1680]
Art	upper	−0.0198	0.0057	−0.1318	0.0019	+0.3165	0.0112	(1200, 1680]
Art	central	−0.0423	0.0015	+0.0488	0.0423 [†]	+0.3893	0.0250	(720, 1200]
Art	lower	−0.0110	0.0054	−0.0954	0.0129	+0.2952	0.0052	(1200, 1680]

and DTT in two. All three portfolio members win somewhere, which establishes complementarity in the sense of Section 3.3; what that is worth is a separate question (Section 6.4).

The signal plan displaces the crossing. Under balanced timing the margin has already changed sign by 1200 veh/h in all four restriction configurations; under arterial-priority timing it is still negative at 1200 veh/h in three of four and changes sign only by 1680 veh/h. The mean margin at 1200 veh/h is +0.5209 under balanced timing against −0.0763 under arterial priority, a difference of 0.5972 cost units, so the displacement is at least one full step of the grid, 480 veh/h; the grid cannot say how much more. The approach is also not uniform: under balanced timing the margin increases with demand in all four cells, whereas under arterial priority it *decreases* from 720 to 1200 veh/h in three of four cells before rising sharply. Restriction location moves the margin very little — the

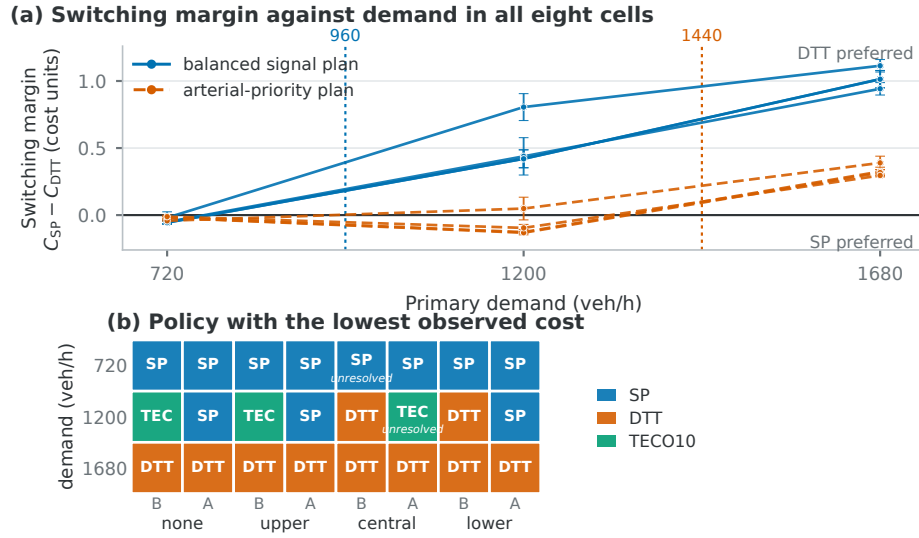


Fig. 3. (a) The switching margin of Eq. (5) in all eight restriction–signal cells; error bars are ± 2 standard errors over three paired seeds and are descriptive, not a significance test. Dotted lines mark the two demand thresholds fitted by the selection rules of Section 6.7. (b) The policy with the lowest observed cost in each context; the two contexts whose margin is not resolved against replication dispersion are marked.

four balanced cells at 1200 veh/h span $+0.4196$ to $+0.8056$, three of them within 0.02 of each other — which foreshadows the feature ablation of Section 6.7.

6.3 Resolution of the Switching Boundary

The experiment brackets the sign change but does not locate it (Section 5.3). What the completed data establish is that the brackets rest on margins large relative to replication dispersion: 22 of the 24 margins in Table 3 satisfy $|m_s| > 2 \text{SE}(m_s)$, the smallest ratio among those being 2.03. The two exceptions are the balanced central-restriction context at 720 veh/h ($m = -0.0195$, $\text{SE} = 0.0220$) and the arterial central-restriction context at 1200 veh/h ($m = +0.0488$, $\text{SE} = 0.0423$). These are exactly the two contexts in which the sign of the margin differs across the three seeds, so the descriptive criterion and the direct seed-by-seed check agree. This also explains the one cell that breaks the signal-plan pattern: the arterial central cell is assigned the (720, 1200] bracket on the strength of a margin whose sign three replications do not agree on, so seven of the eight cells have a resolved bracket and the eighth does not.

Dispersion is not uniform: the standard error of the margin has a median of 0.0057 cost units at 720 veh/h, 0.0334 at 1200 veh/h and 0.0232 at 1680 veh/h, with its maximum, 0.0697, also at 1200 veh/h. Replication noise is largest precisely in the band where the ordering changes — the band in which a decision is hardest

and, as the next subsection shows, the one carrying most of the available decision value.

6.4 Decision Value of Adaptation

Complementarity is established, but its value is modest. Over the 24 contexts the single best fixed policy is DTT, with mean decision regret 0.03018 cost units; fixed SP and fixed TECO10 cost 0.31825 and 0.20757. Under leave-one-context-out the depth-3 selector attains mean regret 0.00606, chooses the lowest-cost policy in 21 of 24 contexts and is within 1 % of the context optimum in 22 of 24, so by Eq. (2) it captures $G = (0.03018 - 0.00606)/0.03018 = 0.799$: 79.9 % of the available decision headroom.

This percentage is a decision statistic and nothing else. It is *not* a 79.9 % reduction in travel time, emissions or any physical quantity. In physical terms the selected policies average 412.17 s and 1114.64 g per vehicle against 425.17 s and 1145.36 g under fixed DTT — 13.00 s (3.06 %) and 30.71 g (2.68 %) lower over the evaluated contexts, with a hindsight bound of 409.25 s and 1106.12 g. The decision statistic says how much of the attainable improvement was realised; the physical figures say how large the attainable improvement was.

The headroom is highly localised (Fig. 4b). At 1680 veh/h DTT is best in all eight contexts, so there is nothing for adaptation to win: the fixed policy has exactly zero mean regret and the entire penalty in that regime belongs to choosing the *wrong* fixed policy, which costs 0.67611 under SP. At 720 veh/h the fixed policy carries 0.03422 and the selector removes all of it; the transition band carries 0.05633, of which the selector removes about two thirds, leaving 0.01818. Weighted by the eight contexts in each regime, 37.8 % of the total headroom sits at low demand, 62.2 % in the transition band and none at high demand.

Two conclusions follow and they point in different directions. Adaptation is worth having, but it is worth much less than choosing the right fixed policy: the difference in mean regret between the best and worst fixed policy, 0.28807 cost units, is nearly ten times the entire headroom adaptation competes for. The first-order operational result is about the fixed choice; adaptation is a second-order refinement concentrated in one demand band.

6.5 Noise and Effect Size

A headroom of 0.03018 should be checked against the dispersion of the benchmark that produced it. Recomputing the whole decision analysis independently within each seed block gives 0.04138, 0.03035 and 0.02959 cost units, a mean of 0.03377 with a standard error of 0.00381 across blocks. The quantity the selector competes for is therefore roughly eight times the dispersion of its own estimate — enough for the comparison above to be meaningful, but not enough to treat its fifth decimal place as stable; the differences that matter are the order-of-magnitude ones, 0.318 against 0.030 against 0.006. At context level the picture is less reassuring in the transition band, where the median absolute margin is 0.2757 cost units against a median standard error of 0.0334 but individual ratios run down to the two

unresolved contexts above, and all three of the selector’s remaining errors fall in that band. The difficulty is concentrated where the data are noisiest — expected for a boundary-detection task, and the main reason the residual 20 % of the headroom is not recovered.

6.6 Mechanistic Interpretation

The reversal has a consistent physical description (Table 4). SP’s route is about 8 % shorter than DTT’s and that advantage is essentially constant across regimes; what changes is the cost of using the shorter corridor. Between the lowest and highest demand, implied in-network mean speed falls from 37.80 to 18.31 km/h for SP, a loss of 51.6 %, against 38.43 to 23.80 km/h for DTT, a loss of 38.1 %, while emission intensity rises from 255.1 to 590.7 g/km for SP but only from 248.5 to 423.2 g/km for DTT. A fixed structural advantage is overtaken by a growing congestion penalty: at 720 veh/h the policies achieve almost identical speeds and queueing, so DTT’s longer route is not compensated, whereas as demand rises the shortest corridor degrades faster than the alternatives and the margin flips sign. Under arterial priority that corridor receives a larger green share, consistent with the later crossing in Table 3.

Table 4. Mechanism decomposition: SP minus DTT, averaged over the eight contexts at each demand level. A negative distance difference means SP’s route is shorter; a positive time difference means SP is slower.

Demand	Δ distance (m)	Δ in-network time (s)	Δ insertion delay (s)	Δ journey time (s)	Δ CO ₂ (g/veh)	Δ stopped (veh-s/veh)
720 veh/h	−250.3	−17.9	0.0	−17.9	−44.7	+6.4
1200 veh/h	−288.0	+84.8	+18.1	+102.8	+322.5	+104.1
1680 veh/h	−277.2	+113.7	+441.2	+554.9	+416.9	+114.3

This is a decomposition, not a causal identification. The experiment varies demand and green split but does not isolate movement-level mechanisms, and the per-run aggregates contain no per-approach delay, stop count or route-share breakdown.

6.7 Context-Aware Selection

Table 5 and Fig. 4a report the ladder under the identical held-out protocol. The central result is the equality of rows R2 and R4: a rule with two free parameters, one demand threshold per signal plan, reproduces the regression tree exactly — same mean regret, same maximum regret, the same 21 contexts chosen correctly. The fitted models explain why: in all 24 folds the tree splits first on demand at 1440 veh/h, then on signal plan, then on demand again at 960 veh/h, while the

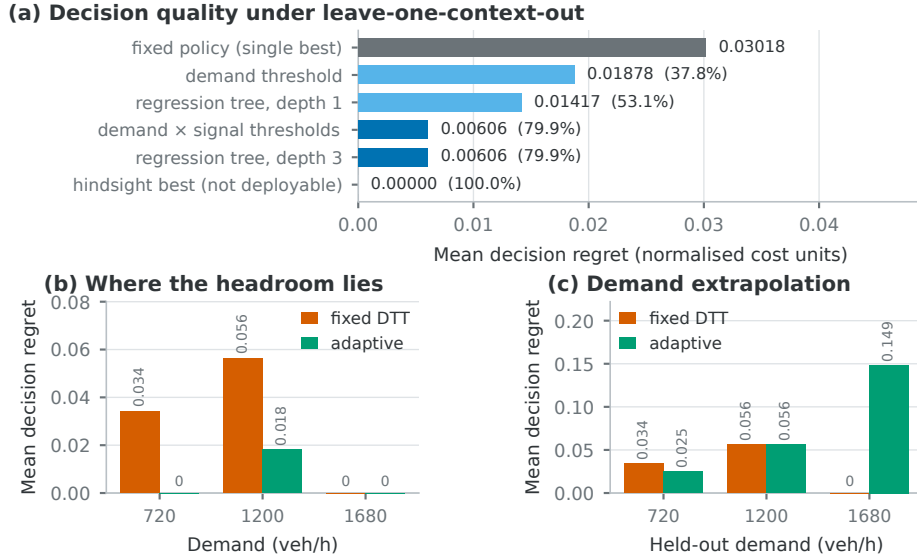


Fig. 4. (a) Mean decision regret of every rule in the ladder under leave-one-context-out, with the percentage of headroom captured. (b) Where the headroom lies: at 1680 veh/h the fixed policy is already optimal. (c) Leave-one-demand-out. When the withheld demand level lies above the fitted support the adaptive rule is substantially worse than the fixed policy it replaces.

explicit rule fits thresholds of 960 veh/h under balanced timing and 1440 veh/h under arterial priority in every fold. The two rules are the same rule.

This bounds what the learned model contributes: the 79.9% is produced by the regime structure of Section 6.2, not by model capacity, and a practitioner would obtain the same decisions from two numbers written on a page. Feature ablation agrees — removing the restriction configuration leaves the result unchanged at 79.9%, whereas removing the signal plan reduces it to 72.9%. The ladder also separates prediction quality from decision quality: the depth-3 model predicts poorly in absolute terms, with held-out mean absolute error of 44.6s in journey time and 82.4g in CO₂ against benchmark means of 542.8s and 1263.5g, yet chooses the lowest-cost policy in 21 of 24 contexts, because errors that preserve the ordering of three policies cost nothing. This is the predict-then-optimize distinction of [9] in a traffic setting, and it means reporting the selector’s regression accuracy would misdescribe its usefulness.

The extrapolation boundary. The leave-one-demand-out experiment reverses the conclusion (Fig. 4c). Pooled over the three folds, mean regret rises to 0.07649, above the 0.03018 of the fixed policy: applied this way the procedure is worse than doing nothing adaptive. The folds are not equivalent. With 1200 veh/h withheld the test point lies between two training levels and the selector reproduces the fixed-policy regret of 0.05633 exactly, gaining nothing; with 720 veh/h withheld it never

Table 5. Selection rules under leave-one-context-out. Everything — reference scales, thresholds, tree and the identity of the best fixed policy — is refitted inside each fold. Mean and max are decision regret in normalised cost units; “lowest cost” counts contexts in which the chosen policy is the observed best; “captured” is $G(\pi)$ of Eq. (2).

Rule	Mean	Max	Lowest cost	Captured
R0 Single best fixed policy	0.03018	0.13176	10/24	—
R1 Demand threshold	0.01878	0.13176	18/24	37.8 %
R3 Regression tree, depth 1	0.01417	0.10079	17/24	53.1 %
R2 Demand $\times$ signal thresholds	0.00606	0.11545	21/24	79.9 %
R4 Regression tree, depth 3	0.00606	0.11545	21/24	79.9 %
R5 Hindsight best (not deployable)	0.00000	0.00000	24/24	100 %

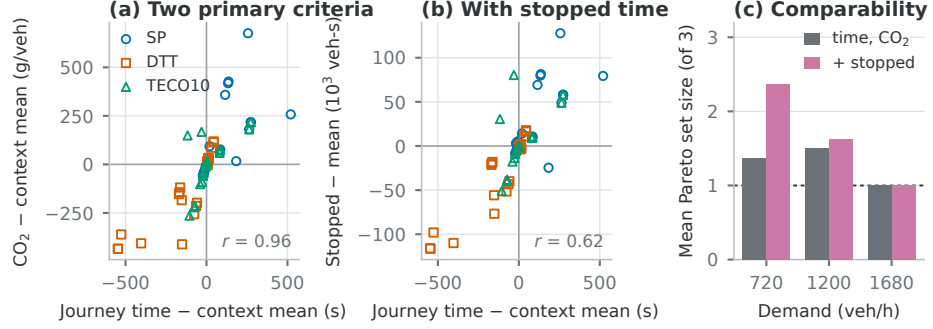


Fig. 5. Multi-criteria structure. (a, b) Policy outcomes expressed as deviations from their own context mean, which isolates the comparison that decides a context; r is the mean within-context correlation across the three policies. (c) Mean number of non-dominated policies per context, with and without the congestion axis.

sees a context in which SP is optimal and incurs 0.02456 against 0.03422, slightly better only because the fixed policy is also wrong there; and with 1680 veh/h withheld the test point lies above the training range, DTT is optimal in all eight withheld contexts, the selector assigns TECO10 to half of them, and it incurs 0.14856 where the fixed policy incurs zero.

The failure is specific and diagnosable rather than general: a tree cannot extrapolate, returning beyond the largest training value of a feature the prediction of its outermost leaf, which here was fitted where TECO10 was competitive. Interpolation inside a represented range is safe in this benchmark, extrapolation beyond it is not, and a model that produces a prediction outside its support is not thereby producing a usable one.

6.8 Multi-Criteria and Pareto Structure

Whether a preference model does any work depends on how independent the criteria are. Within a context — the comparison that determines which policy

wins — the mean correlation between mean journey time and CO₂ across the three policies is 0.96 (Fig. 5a), so the two primary criteria are close to collinear, which is why the weight sweep below changes nothing; the corresponding correlation between journey time and network stopped vehicle-time is 0.62 (Fig. 5b), so the congestion diagnostic carries information the primary criteria do not.

Treating that diagnostic as a third axis changes comparability but not the decision. Under the two primary criteria the mean Pareto set contains 1.29 of the three policies and 17 of 24 contexts have a single non-dominated policy; adding stopped vehicle-time raises the mean set to 1.67 and leaves only 12 such contexts, the effect concentrated at low demand where the mean set size rises from 1.38 to 2.38. At 1680 veh/h it remains 1.00 under both, DTT dominating outright on all three axes. Crucially, the policy chosen by the primary rule is non-dominated in the three-criterion sense in all 24 contexts, so the reported decisions are never Pareto-dominated once congestion is included. An equal-weight additive rule over all three criteria agrees with the primary rule in 22 of 24 contexts — the exceptions being the balanced central-restriction context at 720 veh/h and the balanced upper-restriction context at 1200 veh/h — and ranking by stopped vehicle-time alone agrees in 18 of 24. The congestion diagnostic therefore reduces the *confidence* with which policies can be ordered at low demand, where they differ little on any axis, without altering the regime structure or the decisions. That is also why it was not promoted to a primary criterion: it would have changed the meaning of the reported decisions while changing almost none of them.

6.9 Robustness and Negative Controls

Table 6. Sensitivity checks under the primary protocol. “Captured” is the fraction of headroom recovered; “labels changed” counts contexts in which the preferred policy differs from the primary analysis. All 15 depth/leaf settings capture positive headroom. The two negative controls are described in the text. *On the in-network scale the headroom is 0.03305 rather than 0.03018.

Variation	Captured	Labels changed
Primary analysis (depth 3, leaf 2, $w = 0.5$)	79.9 %	—
Depth $\in \{1, 2, 3, 4, \infty\} \times$ leaf $\in \{1, 2, 3\}$	50.9–79.9 %	—
Weight w_{time} swept $0 \rightarrow 1$	78.1–81.7 %	0 of 24 decisions
Seed-block hold-out, three blocks	60.8, 67.7, 90.2 %	—
Training-only scoring scales	79.8 %	—
In-network time replaces journey time	80.1 %*	0 of 24
95th percentile replaces the mean	—	2 of 24
Route distance as a third criterion	—	0 of 24
Stopped vehicle-time as a third criterion	—	2 of 24

Table 6 collects the checks of Section 5.4; the held-out advantage survives all of them. Both negative controls fire, and both constrain the paper’s own claims.

The emission-forecast control is the reason ECO is not in the decision portfolio. Its aggregate accuracy is acceptable — selected-path emission WAPE of 14.71 % against a 25 % criterion and signed bias -0.32 % against a 10 % criterion — yet in a fixed-replay probe over all 29 admissible routes it identified the best route in only 3 of 8 panels, with a mean relative selection loss of 21.62 % and a worst case of 78.16 %. Good average forecast accuracy did not produce good route choices, the accuracy–decision gap of Section 6.7 in the opposite direction. ECO is also outcome-identical to SP in 21 of 24 contexts here. Neither observation is evidence that eco-routing is ineffective in general; both are statements about this estimator in this environment.

The preference-model control is a negative result about this study’s own apparatus: sweeping the time weight from 0 to 1 leaves every selector decision unchanged and changes the hindsight winner in a single context, so the multi-criteria layer is doing almost no work here. Adding further multi-criteria methods would produce agreement among methods rather than evidence about the traffic system, and we therefore report the additive model as a transparent definition of “preferred” rather than as a contribution.

6.10 Implications and Limitations

The three questions of Section 1 receive different answers, and the differences are the point: the preferred policy changes with regime, the decision value of that change is real but small and localised, and the recovery is deployable only inside the represented regimes. For practice this suggests a modest design — commit first to the right fixed policy, which is where most of the available cost sits, then add adaptation as two demand thresholds conditioned on the active signal plan, with a declared operating domain monitored against the demand range used to fit it and a fixed-policy fallback outside it.

The limitations are concrete. *External validity*: one synthetic topology, one primary origin–destination structure, specified rather than calibrated lane capacities and a homogeneous fleet, so the brackets, headroom and capture percentages are benchmark-specific quantities. *Resolution*: 24 contexts, three replications and a demand grid of 480 veh/h; the experiment brackets the preference change rather than locating it, the runs that would locate it were not available (Section 5.3), two of the 24 margins are unresolved, and leave-one-context-out estimates procedure quality on folds that share most of their training contexts. *Behaviour*: restrictions are announced, compliance is complete and routes are assigned once at departure, so nothing here speaks to heterogeneous compliance, en-route control, or the feedback that would arise if a large share of traffic followed the same guidance. *Criteria and measurement*: the two primary criteria are nearly collinear here, so the preference model is close to inert; journey time includes insertion delay while in-network CO₂ does not, which leaves the ordering unaffected but not the absolute values at the highest demand; and the congestion diagnostic is measured over all vehicles rather than the cohort. *Mechanism*: the per-run aggregates

support a consistent decomposition of the reversal but not a causal attribution to particular movements or signal phases. *Scope*: the review is targeted rather than systematic, and the paper accordingly claims no priority.

7 Conclusion

This study treats the choice of a routing policy as a context-dependent decision problem and evaluates it on a controlled signalised benchmark of 432 completed SUMO runs over 24 contexts. Shortest-distance routing has the lowest observed cost in all eight low-demand contexts and dynamic travel-time routing in all eight high-demand contexts, with the crossing displaced by at least one demand step under arterial-priority timing, and all three portfolio policies win somewhere, so the portfolio is complementary.

Complementarity, however, is not decision value. The single best fixed policy leaves only 0.03018 normalised cost units of headroom, of which a leave-one-context-out selector recovers 79.9% — a share of attainable decision quality, not a physical improvement rate; the corresponding physical differences are 13.00 s and 30.71 g per vehicle, while choosing the worst rather than the best fixed policy costs 0.28807 units. A two-threshold rule on demand and signal plan matches the fitted regression tree exactly, so the recoverable value comes from the regime structure and not from model capacity.

Decision value is in turn not deployability. When an entire demand regime is withheld, pooled regret rises to 0.07649, above the fixed-policy benchmark, and in the fold that extrapolates above the training range the adaptive rule incurs 0.14856 where the fixed policy incurs nothing. The recommendation is therefore conditional and operationally specific: context-dependent routing-policy selection is worth deploying only within an explicitly declared regime domain, implemented as a small number of interpretable thresholds, with a fixed-policy fallback whenever the operating context leaves it. Establishing where inside the observed brackets the preference actually changes requires intermediate-demand simulations on this same network, which remain to be run.

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